

Keeping the lights on

Name: _____ Class: _____ Date: _____

Power is how fast energy is supplied, in megawatts (MW). **Energy** is power × time, in megawatt-hours (MWh).

energy (MWh) = power (MW) × time (hours) e.g. 2 MW for 3 hours = 6 MWh

Part A · Power and energy

A1 A solar farm makes 4 MW in full sun. How much energy does it make in 3 hours of full sun? [1]

A2 On a sunny summer day one solar farm makes 40.8 MWh. The town uses 120 MWh in a day.

a) How many solar farms make at least as much energy as the town uses in a day? [1]

b) Explain why these solar farms on their own would still leave the town in the dark at night. [2]

A3 From 8 pm to 6 am the town uses 40 MWh. Each battery stores 8 MWh.

a) What is the smallest number of batteries that could store 40 MWh? [1]

b) Batteries keep 90% of the energy that goes into them. How much solar energy must go in to get 40 MWh out? [2]

A4 In **Evening Rush**, the town needs 14 MW at 6 pm. The sun has set and the wind turbine gives only 0.15 MW. Each battery can give out at most 2 MW.

a) What is the fewest batteries that can give out the rest, all at once? [2]

b) How much energy can that many batteries store altogether? [1]

c) The 6 pm hour needs only 13.85 MWh from the batteries. So why do you need this many? [2]

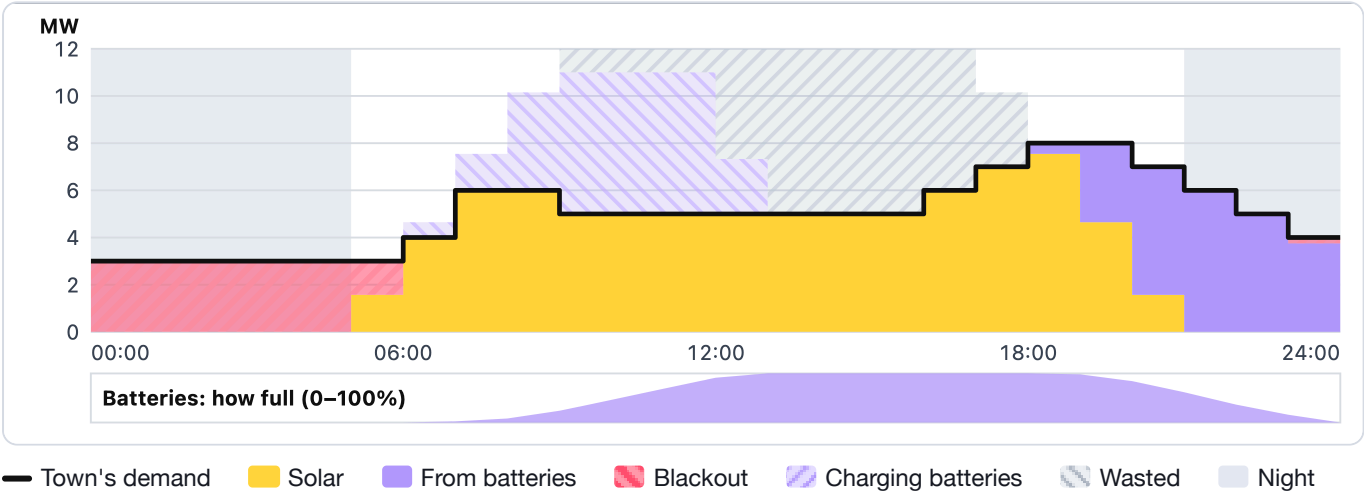
Part B • Play levels 1 to 4

Before you press **Run the day**, predict what will happen. Record your best design for each level.

Level	What I built	Cost	Stars	What went wrong, and how I fixed it
1		£		
2		£		
3		£		
4		£		

Part C • Read the chart

This chart is from **After Dark**, with **4 solar farms and 3 batteries**. Below the black line is where the town's power came from; above it is spare solar power. The strip underneath shows how full the batteries are. The day repeats, so the night on the left of the chart (midnight to 6 am) follows on from the evening on the right.



C1 At what time do the batteries start charging? [1]

C2 At what time are the batteries full? [1]

C3 From 09:00, some solar power is wasted (grey) even though the batteries are not full yet. Why? [2]

C4 When does the blackout start, and when does it end? [2]

C5 This design wastes 57 MWh of solar energy but is still 17 MWh short. Should you add more solar farms or more batteries? Explain your answer. [3]

Part D • Explain

D1 Wind turbines cost more than solar farms (£40 a day against £30). Yet the cheapest design for **Windy Hill** uses two wind turbines and costs less than the cheapest design for the same town in **After Dark** (£230 against £300). Explain why. [4]

Designing a low-carbon grid

Name: _____ Class: _____ Date: _____

Capacity factor = energy produced ÷ (rated power × time). **Efficiency** = energy out ÷ energy in.
 Batteries: 2 MW, 8 MWh, keep 90%, £30 a day. Pumped hydro: 4 MW, 60 MWh, keeps 75%, £60 a day.
 Gas plant: up to 4 MW, £10 a day plus £2 per MWh of gas, releases 0.4 t of CO₂ per MWh.

1 Capacity factor.

a) On Windy Hill's breezy day, a 3 MW turbine on a hilltop produces 30 MWh in 24 hours. Calculate its capacity factor. [2]

b) The same turbine in a field produces 15 MWh. Calculate its capacity factor, and suggest why engineers measure the wind before choosing a site. [2]

c) In Grey Skies, a 4 MW solar farm produces 33.9 MWh on sunny Monday and 8.9 MWh on grey Tuesday. Calculate both capacity factors. [2]

2 Cost per MWh. A solar farm costs £30 a day.

a) Work out its cost per MWh on Monday and on Tuesday (use 1c). [2]

b) A battery costs £30 a day and can deliver at most 8 MWh in a day. What is the lowest cost per MWh it adds to energy that is stored? [1]

c) Use your answers to explain why the cheapest Grey Skies design puts wind turbines on all three hilltops. [3]

3 CO₂ in Backup Plan. The CO₂ limit is 4 t for the day.

a) What is the most energy that may come from gas? [1]

b) With no CO₂ limit, the cheapest design (3 wind turbines and 3 gas plants) costs £317 a day and burns 83.7 MWh of gas. How much CO₂ does it release? [1]

c) With the limit, the cheapest design costs £409 and releases 1.9 t. By what percentage does the limit raise the cost, and by what percentage does it cut CO₂? [2]

d) Work out the extra cost per tonne of CO₂ avoided. [2]

4 Storage.

a) Pumped hydro keeps 75% of the energy pumped in. To deliver 45 MWh, how much energy must go in? [1]

b) How much would have to go into batteries to deliver the same 45 MWh? [1]

c) Work out the cost per MWh of storage capacity for pumped hydro and for batteries. [2]

d) The cheapest Still Week design still includes one battery and one gas plant. The town's demand peaks at 12 MW. Suggest why. [2]

5 Demand-side response (Smart Timing).

a) Bus charging needs 4 MW for 3 hours. How much energy is that? [1]

b) If the buses charge from 19:00 to 22:00, after sunset, all of that energy must come out of batteries. How much solar energy has to go into the batteries, and how many 8 MWh batteries are needed to store the 12 MWh the buses use? [3]

c) The three flexible jobs use 32 MWh, about a fifth of the town's 150 MWh. Moving them into the morning and midday cuts the cheapest design from £390 to £280 a day. Suggest two real ways to get people or businesses to move when they use electricity. [2]

6 A still, grey week (Still Week). The town needs 171 MWh every day. In the cheapest design, five turbines at sea make 34.7 MWh on still Tuesday, and there are no solar farms.

a) How much energy must come from storage and gas on Tuesday? [1]

b) The CO₂ limit (4 t) allows 10 MWh of gas over all three days. At least how much must come from storage on Tuesday? [1]

c) Without pumped hydro, the cheapest design needs 16 batteries and costs £696 a day, against £366 with it. Explain why long-duration storage matters for a renewable grid. [2]

7 Evaluate. A town wants to run on renewable energy with almost no CO₂. Evaluate the ways of keeping the lights on when the wind and sun are weak: batteries, pumped hydro, demand-side response and gas backup. Use evidence from Grid Manager. [6]

Continue on lined paper if you need more space.